STANDARD OPERATING PROCEDURE			P-3810	
	SARPY COUNTY SHERIFF'S OFFICE PAPILLION * NEBRASKA **GREG LONDON** **SHERIFF**	DATE ISSUED 06/01/06	DATE REVISED **03/19/24**	DATE REVIEWED 02/02/25
		APPROVED: *GREG LONDON*		
CHAPTER TITLE ROAD PATROL		RESOURCES		
SUBJECT RADAR/LIDAR		CROSS REFERENCE P-3200 PATROL UNIT		

I. Purpose:

The purpose of this policy is to establish procedures for using RADAR (Radio Detection and Ranging) and LIDAR (Light Detection and Ranging) speed detection equipment.

II. Policy:

The Sarpy County Sheriff's Office seeks to ensure the safety of motorists traveling within Sarpy County. Excessive speed has long been recognized as a major contributing factor to death and injury on the roadways of America. The control of excessive speed is of considerable interest to the Sheriff's Office. The sole purpose of the RADAR and LIDAR equipment is to detect excessive speed. Scientific testing has shown that RADAR and LIDAR instruments used in traffic enforcement are reliable when properly installed and operated by skilled and knowledgeable individuals.

III. Procedure:

A. RADAR-Deputy Responsibilities: Only deputies that are trained and qualified to operate RADAR equipment shall do so.

1. Accuracy Check: Prior to utilizing a RADAR unit, the deputy shall verify the equipment is functioning properly and is providing accurate speed-readings. Deputies shall perform this test at least twice during a shift (the beginning and the end) when the RADAR is operated. Deputies shall perform the following accuracy tests:

a) Internal Circuit Test: Deputies shall perform an internal circuit test. This varies by manufacturer but usually consists of depressing the "test" button and ensuring a readout of "PASS" or "OK" in the display window.

b) External Tuning Fork Test: Deputies shall perform an external tuning fork test in moving and stationary modes on all antennas. In stationary

mode, the target window should display the speed stamped on the tuning fork. In moving mode, both tuning forks are used simultaneously and the target window should display the highest speed stamped on the tuning fork minus the lower speed stamped on the second tuning fork; the patrol window should display the lower speed. All speeds should be within (1) M.P.H. of the listed speed on the tuning fork. Tuning forks shall not be struck against hard objects or metal that can damage or cause dents to the forks. Tuning forks shall be struck against softer materials such as the plastic butt plate of a magazine or a padded steering wheel. Tuning forks are assigned and serialized to each individual radar unit. Turning forks shall not be used to test other radar units.

c) **Patrol Unit Speed Verification:** With the RADAR unit in moving mode, the deputy shall verify the patrol unit speedometer and the patrol unit display are within (1) M.P.H.

2. **Malfunction of RADAR unit:** If a RADAR unit fails the accuracy check, lacks a Doppler tone, or is functioning unreliably, the deputy shall immediately discontinue using the unit and notify his or her shift supervisor. The deputy shall then complete a vehicle maintenance request (see P-3200 Patrol Unit, Section III, A, 4).

3. **Location:** Deputies should select a proper and safe location to monitor traffic.

4. **Tracking History:** Deputies shall ensure that all elements of the tracking history have been accounted for prior to writing a citation for speeding using the RADAR unit:

a) **Visual Observation:**

(1) Identify target vehicle.
(2) Estimate the target vehicle's speed.
(3) Observe the environment surrounding the vehicle.

b) **Audio Confirmation:**

(1) A Doppler tone with a pitch matching the target vehicle's speed is audible (the faster the target vehicle's speed the higher the pitch of the Doppler tone).
(2) The Doppler tone should be clear.

c) **RADAR verification:**

(1) The RADAR displays a stable readout of the targets speed (speed displays for approximately 3 to 5 seconds).

(2) Patrol unit's speedometer matches the patrol unit display on the RADAR unit (moving mode ONLY).

(3) The target vehicle's speed **may** be locked (optional).

B. LIDAR-Deputy Responsibilities: Only deputies that are trained and qualified to operate LIDAR equipment shall do so.

1. Accuracy Check: Prior to utilizing a LIDAR unit, the deputy shall verify the equipment is functioning properly and is providing accurate speed-readings. Deputies shall perform this test at least twice during a shift (the beginning and the end) when the LIDAR is operated. Deputies shall perform the following accuracy tests:

a) Internal Self-Test: The deputy shall depress the "TEST" button and ensure a reading of "PASS" is displayed.

b) Scope Alignment: The deputy shall ensure the scope mounted to the LIDAR unit is accurate in windage (left to right) and elevation (up and down). The deputy shall listen for the presence and absence of the audible tone when aiming at a fixed object and then moving off that fixed object.

c) Delta Distance Test: The deputy shall use the LIDAR unit to determine the distance to two pre-measured points. The LIDAR unit must display the known distances within 1 foot. One distance should be less than 200 feet and the other greater than 200 feet and the distances should be measured using a tape measurer.

d) Patrol Unit Speed Verification (optional): The deputy **may** arrange for a second patrol unit to approach his or her location at a pre-arranged speed. The deputy shall ensure that the target's speed-reading matches the patrol unit's speedometer (by use of radio communications) within 1 M.P.H. This test is optional.

2. LIDAR Use: The LIDAR unit shall only be used under the following conditions:

a) The LIDAR unit shall not be used during weather conditions that include precipitation (e.g. rain, snow, sleet, etc.).

b) The LIDAR unit shall not be used to obtain speed-readings through car windows.

c) The LIDAR unit shall not be used in a moving vehicle.

d) Citations shall include the distance at which the speed-reading was obtained and name and badge of LIDAR operator if other than the deputy issuing the citation.

e) The LIDAR operator must take into account additional vehicle traffic as the distance between the operator and the target vehicle increases past 1000 feet. At distances greater than 1000 feet, the operator shall make note of any other vehicles in the area and any relationship between the target vehicle and non-target vehicles.

3. **Malfunction of LIDAR unit:** If a LIDAR unit fails the accuracy check, lacks an audible tone, or is functioning unreliably, the deputy shall immediately discontinue using the unit and notify the Patrol Division Commander.

4. **Location:** Deputies should select a proper and safe location to monitor traffic.

5. **Tracking History:** Deputies shall ensure that all elements of the tracking history have been accounted for prior to writing a citation for speeding using the LIDAR unit:

a) **Visual Observation:**

(1) Identify target vehicle.
(2) Estimate the target vehicle's speed.
(3) Verify the distance to the target vehicle.
(4) Observe the environment surrounding the vehicle.
(5) Place the scope's red dot aim point on the front of the target vehicle.

b) **Audio Confirmation:**

(1) The LIDAR provides an audible tone advising when the LIDAR is transmitting and a single tone indicating a distance and speed are locked.

c) **LIDAR verification:**

(1) The LIDAR displays a stable readout of the target vehicle's speed (speed displays for approximately 3 to 5 seconds).
(2) The target vehicle's speed and distance will be locked automatically by the LIDAR unit.